# COHOMOLOGY OVER BASIS OPENS COMPUTES ON COHOMOLOGY SHEAVES

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*Editorial note.* All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted span cites the lemmas recorded in assembly.external_references with canonical tags as navigation pointers.

*Allowed standard prerequisites (editorial).*

- Derived limit of truncations. Cohomology of Sheaves, Lemma cohomology-lemma-is-limit-dimension. K = R lim tau_{>=-n}K under the per-point vanishing bound with d = 0.
- Spectral sequence of a bounded-below complex. Cohomology of Sheaves, Example cohomology-example-spectral-sequence. Computes H^q(U, tau_{>=-n}K) from the cohomology sheaves.

*Source reference.* Original lemma 0D64: sources/stacks-chow/tags.txt, source lines 16509–16509. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original equation 0BKR: sources/stacks-chow/tags.txt, source lines 14569–14569. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 0BKM: sources/stacks-chow/tags.txt, source lines 14565–14565. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

**Lemma 0.1.** *Let $(X, \mathcal{O}_X)$ be a ringed space. Let $K$ be an object of $D(\mathcal{O}_X)$. Let $\mathcal{B}$ be a set of opens of $X$. Assume*

(1) *every open of $X$ has a covering whose members are elements of $\mathcal{B}$,*
(2) *$H^p(U, H^q(K)) = 0$ for all $p > 0$, $q \in \mathbf{Z}$, and $U \in \mathcal{B}$.*

*Then $H^q(U, K) = H^0(U, H^q(K))$ for $q \in \mathbf{Z}$ and $U \in \mathcal{B}$.*

*Proof.* Observe that $K = R\lim \tau_{\geq -n} K$ by Lemma 0D64 with $d = 0$. Let $U \in \mathcal{B}$. By Equation (0BKR) we get a short exact sequence

$$0 \to R^1 \lim H^{q-1}(U, \tau_{\geq -n} K) \to H^q(U, K) \to \lim H^q(U, \tau_{\geq -n} K) \to 0$$

Condition (2) implies $H^q(U, \tau_{\geq -n} K) = H^0(U, H^q(\tau_{\geq -n} K))$ for all $q$ by using the spectral sequence of Example 0BKM. The spectral sequence converges because $\tau_{\geq -n} K$ is bounded below. If $n > -q$ then we have $H^q(\tau_{\geq -n} K) = H^q(K)$. Thus the systems on the left and the right of the displayed short exact sequence are eventually constant with values $H^0(U, H^{q-1}(K))$ and $H^0(U, H^q(K))$. The lemma follows. □

*Date*: stacks-project:cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.